Question #1 of 60

Question ID: 1674564

Which of the following statements regarding the Black-Litterman model is *most accurate*?

A) Implied returns from reverse optimization are view adjusted and then used as inputs to mean variance optimization.

B) Implied returns from reverse optimization are used as inputs to mean variance optimization (MVO).

C) The Black-Litterman model starts with the global market portfolio's weights, which are assumed to be optimal.

Explanation

The Black-Litterman model uses the implied returns from the output of reverse optimization and then view adjusts those returns according to the beliefs of the investor, with the view adjusted returns used as inputs to MVO. *Reverse optimization* starts with the global market portfolio's weights, which are assumed to be optimal.

(Module 4.1, LOS 4.a)

Question #2 of 60

Question ID: 1580688

Daniel Roe and Loretta Morgan are both potential new clients of Grachek Investment Advisors. A summary of Ellen Grachek's notes concerning the potential clients are as follows:

- Roe stated that he wants to have a positive return on his $500,000 portfolio at all times, and that his required before-tax return is 5%. On a risk aversion questionnaire, Roe scored an 8, with 10 indicating the highest risk aversion.
- Morgan indicated that her $1,000,000 portfolio must generate a 2% return each year to meet her living expenses without making any withdrawals from the portfolio's principal. On a risk aversion questionnaire, Morgan scored a 3, with 10 indicating the highest risk aversion.

Grachek Investment Advisors has four model portfolios that they use for each client. Characteristics for each portfolio are identified as follows:

Portfolio	Expected Return	Standard Deviation
A	5.5%	7%
B	6.0%	8%
C	6.5%	10%
D	8.0%	15%

After reviewing her notes and making some calculations, Grachek makes the following statements regarding each client:

Statement 1:	Based on a utility-adjusted return of 2.54%, Portfolio B would be the best choice for Roe.
Statement 2:	Using Roy's safety-first measure, Portfolio D generates a score of 0.40, and would be the worst choice of the four for Morgan's portfolio.

Regarding Grachek's statements:

A) Statement 1 is incorrect; Statement 2 is correct.

B) Statement 1 is correct; Statement 2 is correct.

C) Statement 1 is incorrect; Statement 2 is incorrect.

Explanation

Grachek's statement regarding Roe is based on utility-adjusted return.

Utility-adjusted return = $U_p = R_p - 0.005(A)(\sigma^2)$

$U_A = 5.5\% - 0.005\ (8)\ (7\%^2) = 3.54\%$

$U_B = 6.0\% - 0.005\ (8)\ (8\%^2) = 3.44\%$

$U_C = 6.5\% - 0.005\ (8)\ (10\%^2) = 2.50\%$

$U_D = 8.0\% - 0.005\ (8)\ (15\%^2) = -1.00\%$

Based on the calculations, Portfolio A would be the best choice for Roe with a utility-adjusted return of 3.54%. Statement 1 is incorrect.

Grachek's statement regarding Morgan is based on Roy's safety-first measure. Morgan indicated that she has a minimum 2% return.

Roy's safety-first measure is calculated as: $SF = (R_P - R_{MIN}) / \sigma_P$

$SF_A = (5.5 - 2) / 7 = 0.50$

$SF_B = (6.0 - 2) / 8 = 0.50$

$SF_C = (6.5 - 2) / 10 = 0.45$

$SF_D = (8.0 - 2) / 15 = 0.40$

Based on the calculations, Portfolio D would be the worst choice for Morgan's portfolio. Statement 2 is correct.

(Module 4.3, LOS 4.b)

Question #3 of 60

Question ID: 1587532

A portfolio manager expects that transaction costs will rise next year and that the correlation of equities with the rest of the portfolio will rise. What is the net impact of those changes on the portfolio's optimal rebalancing corridor width?

A) They will result in a wider corridor width.

B) They will result in a narrower corridor width.

C) They will result in an uncertain impact on corridor width.

Explanation

The rise in transaction costs and the rise in asset class correlation with the rest of the portfolio will result in an overall wider optimal portfolio width:

- Higher transaction costs allow for a wider corridor width because the asset mix would have to change more to justify the higher cost of rebalancing.
- Higher correlation of equities with the rest of the portfolio will lead to a wider corridor because the portfolio will move more with the asset class, and the allocations will tend to stray more slowly from the target.

(Module 4.8, LOS 4.o)

Question #4 of 60

Question ID: 1580693

Melissa Brown, an analyst with Mollette Capital Advisors (Mollette), is reviewing the client profile of Karrie Jones. Mollette manages all of Jones's investment assets; however, because Brown is new to the firm, she has never met Jones. She does know, however, that Jones's asset allocation is appropriate given her age and investment policy statement. The allocation of Jones's portfolio is shown as follows:

Asset Class	Allocation (%)
Cash	5%
Intermediate-term Treasury bonds	0%
High-quality corporate bonds	5%
U.S. equities	60%
International equities	30%

Given Jones's asset allocation, which of the following conclusions about Jones is *most accurate*?

A) Jones has a large amount of financial capital. ✗

B) Jones has a low risk tolerance. ✗

C) Jones's human capital makes up the bulk of her portfolio. ✓

Explanation

With only 10% in fixed assets and the rest in equities, Jones has a very aggressive portfolio. At a young age, an aggressive portfolio like this makes sense as the individual will have a high allocation to human capital (future stream of income from working). At this stage, the allocation to human capital will be much larger than the allocation to financial capital (investment portfolio). Therefore, the investment portfolio should be invested in riskier, high-return assets. Note that Jones likely has a high risk tolerance given the aggressive portfolio.

(Module 4.4, LOS 4.c)

Question #5 of 60

Question ID: 1551694

Which of the following asset classes is *least likely* to require a liquidity return premium?

A) Direct real estate.

B) Infrastructure.

C) Commodities.

Explanation

Commodities are not generally thought of as a less liquid asset class requiring a liquidity return premium to compensate the investor for the additional liquidity risk. On the other hand, asset classes such as direct real estate and infrastructure would be considered less liquid and would require a liquidity return premium.

(Module 4.4, LOS 4.f)

Question #6 of 60

Question ID: 1580698

Which of these functions is *least likely* to be addressed with Monte Carlo simulation?

A) Assisting individual investors in identifying risk tolerance levels.

B) Rebalancing and taxes of a portfolio in a multiperiod framework.

C) Transforming a dataset with a nonnormal distribution to a normal distribution.

Explanation

Transforming datasets from nonnormal distributions to normal distributions is not addressed by Monte Carlo simulation. Multiperiod analysis of rebalancing and taxes within a portfolio is difficult to analyze mathematically, but it is relatively straightforward using Monte Carlo simulation. Monte Carlo simulation can be useful in illustrating the range and likelihood of possible outcomes given various assumptions of how much risk to take on. This provides the investor with more information to decide on a risk tolerance level with which the investor is comfortable.

(Module 4.4, LOS 4.e)

Question #7 of 60

Question ID: 1551722

A pension fund has pension assets that significantly exceed the value of pension liabilities. The fund wants to minimize the risk that the pension plan will become underfunded. Which of the following liability-relative approaches is *most appropriate* for the fund?

A) Integrated asset-liability approach. ✗

B) Surplus optimization. ✗

C) Two-portfolio (return-seeking) approach. ✓

Explanation

The two-portfolio approach is most appropriate for the pension fund because it allows the fund to create an asset portfolio that hedges its liabilities, while separately creating a portfolio that manages the remaining assets using mean-variance optimization (MVO) under a return-seeking approach. Should the funded status of the plan deteriorate in the future, the manager could reduce allocation to the return-seeking portfolio and increase allocation to the hedging portfolio.

Surplus optimization is an extension of MVO that manages the portfolio surplus against the surplus volatility.

An integrated asset-liability approach jointly optimizes assets and liabilities, typically against future changes in multiple factors.

(Module 4.6, LOS 4.k)

Question #8 of 60

Question ID: 1580685

The following information is available regarding corner portfolios from an efficient frontier.

Corner Portfolio	Expected Return	Exp. Std. Dev.	Sharpe Ratio	Asset Class Weights			
				1	2	3	4
1	9.90%	6.60%	1.197	0.00%	11.10%	88.90%	0.00%
2	10.06%	6.64%	1.214	0.00%	14.90%	85.10%	0.00%
3	12.10%	9.58%	1.054	55.22%	0.00%	44.78%	0.00%
4	14.23%	14.74%	0.830	100.00%	0.00%	0.00%	0.00%

An investor has a spending rate of 6%. If inflation is expected to be 4.50% annually and the cost of earning investment returns is 0.5%, what would be the utility of the portfolio that will, at a minimum, satisfy the investor's goals of capital preservation in real terms with a risk aversion value of 4?

A) 0.120.

B) 0.078.

C) 0.099.

Explanation

$r = (1 + s)(1 + i)(1 + c) - 1 = (1.06)(1.045)(1.005) - 1 = 11.32\%$

This portfolio would lay between corner Portfolios 2 and 3. Let *w* denote the weight of corner Portfolio 2. We solve for *w* in the following equation:

$11.32 = (10.06)(w) + (12.10)(1 - w)$

$w = 0.38$

Approximate standard deviation of portfolio = (0.38)(6.64) + (0.62)(9.58) = 8.46%

$R_Z = 4$

$U = E(R) - 0.5(R_Z)(\sigma^2) = 0.1132 - 0.5(4)(0.0846)^2 = 0.099.$

(Module 4.3, LOS 4.b)

Question #9 of 60

Question ID: 1587521

Which of the following statements regarding liabilities in the context of the asset allocation decision is correct?

A) A university endowment contribution is an example of a legal liability.

B) The pension liability associated with a defined pension plan is an example of a fixed liability.

C) A small pension liability in relation to the size of the sponsoring organization should be ignored.

Explanation

A small liability in relation to the size of the sponsoring organization can usually be ignored, as its effect on the optimal asset allocation is minimal.

The pension liability associated with a defined pension plan is an example of a *contingent* liability because the cash flows depend on uncertain future events.

University endowment contributions are an example of *quasi-legal* liabilities, which are not legal obligations, but are cash outflows expected to occur in the future and are essential to the mission of the institution.

(Module 4.6, LOS 4.j)

Question #10 of 60

Question ID: 1551721

Which of the following statements regarding the two-portfolio (or the hedging/return-seeking) approach to liability-relative asset allocation is correct?

A) The asset allocation decisions regarding the two portfolios are made in conjunction with each other.

B) If the funding ratio is greater than 1, then it becomes difficult to create a hedging portfolio that completely hedges the liabilities.

C) The hedging portfolio could be constructed using cash flow matching.

Explanation

The hedging portfolio could be constructed using various techniques such as cash flow matching, duration matching, and immunization.

If the funding ratio is *less* than 1, then it becomes difficult to create a hedging portfolio that completely hedges the liabilities.

The two-portfolio approach has the distinctive feature that the composition of the liabilities is already in place when the asset allocation decisions are made, so the two decisions are made *independently*.

(Module 4.6, LOS 4.k)

Question #11 of 60

Question ID: 1587518

Which of the following methods is the *most appropriate* way of incorporating client risk preferences into asset allocations?

A) Specify additional constraints. ✓

B) Specify a diversification objective. ✗

C) Specify a risk tolerance factor. ✗

Explanation

One way to incorporate investor risk preferences into an asset allocation decision is to specify additional constraints, such as setting limits on allocations to risky asset classes or setting a ceiling on portfolio risk.

Specifying a risk *aversion* factor (not risk tolerance factor) is another way to incorporate investor risk preferences into an asset allocation decision.

There is no such thing as a diversification objective, per se, in the reading.

(Module 4.5, LOS 4.h)

Question #12 of 60

Question ID: 1587516

The following information is available regarding corner portfolios from an efficient frontier:

Corner Portfolio	Expected Return	Exp. Std. Dev.	Sharpe Ratio	Asset Class Weights				
				1	2	3	4	5
1	14.00%	18.00%	0.639	0.00%	0.00%	100.00%	0.00%	0.00%
2	13.66%	16.03%	0.696	0.00%	0.00%	86.36%	0%	14.00%
3	13.02%	13.58%	0.775	21.69%	0.00%	56.56%	0.00%	21.76%
4	12.79%	13.00%	0.792	21.48%	0.00%	52.01%	5.24%	21.27%
5	10.54%	8.14%	0.988	9.40%	51.30%	26.55%	0.00%	12.76%
6	8.70%	6.32%	0.981	0.00%	89.65%	4.67%	0.00%	5.68%

The following portfolios are under consideration by an investor:

Portfolio	Expected Return
A	11.0%
B	13.5%

For an investor with a risk-aversion of 6, which portfolio would have the highest utility?

A) Portfolio A with a utility of 0.092. ✗

B) Portfolio A with a utility of 0.085. ✓

C) Portfolio B with a utility of 0.115. ✗

Explanation

For Portfolios A and B, we first need the approximate standard deviation.

Portfolio A with an expected return of 11% lies between Corner Portfolios 4 and 5. Let *w* denote the weight of Corner Portfolio 5, we solve for *w* in the following equation:

$11 = (10.54)(w) + (12.79)(1 - w)$

$w = 0.80$

Approximate standard deviation of Portfolio A = (0.80)(8.14) + (0.20)(13) = 9.11.

Similarly, Portfolio B with an expected return of 13.50% lies between Corner Portfolios 2 and 3. Let *w* denote the weight of Corner Portfolio 2, we solve for *w* in the following equation:

$13.50 = (13.66)(w) + (13.02)(1 - w)$

$w = 0.75$

Approximate standard deviation of Portfolio B = (0.75)(16.03) + (0.25)(13.58) = 15.418.

$R_Z = 6$

$U_A = E(R_A) - 0.5(R_Z)(\sigma^2_A) = 0.11 - 0.5(6)(0.0911)^2 = 0.085$

$U_B = E(R_B) - 0.5(R_Z)(\sigma^2_B) = 0.135 - 0.5(6)(0.154)^2 = 0.064$

(Module 4.5, LOS 4.h)

Question #13 of 60

Question ID: 1587523

Todd Zattau is the chief financial officer for the Crandall Steel Company, a mature U.S. steel processing company. The company provides a traditional defined benefit pension plan to all of its employees. The plan covers 5,000 employees, and the average age of workers who will eventually collect benefits is 52. Approximately 45% of the plan's participants are now retired and are receiving benefits. Zattau has hired Kara Rittenhouse, a financial advisor, to help him construct an IPS for the plan as well as recommend revisions to the plan's current investment allocation.

Zattau's progress on the IPS so far is shown here:

Return Objective	The discount rate applied to liabilities is 6.5%. Desired level of returns is 7.2%.
Risk Tolerance	?
Liquidity	?
Time Horizon	Company is a going concern, and new employees are still being added to the defined benefit plan, so the actual time horizon of the plan is infinite. However, the high percentage of retired participants and older workforce reduces the effective time horizon of the plan considerably.
Legal/Regulatory	Plan is subject to ERISA requirements.
Taxes	None
Unique Considerations	Plan is currently underfunded by 4%.

The current investment allocation for the plan is shown here:

Asset Class	Allocation (%)	Expected Return
Cash	3%	3.0%
Intermediate-term Treasury bonds	25%	5.0%
High-quality corporate bonds	32%	5.5%
U.S. equities	10%	8.5%
International equities	10%	10.0%
Venture capital	15%	19.0%

Based on the information provided, what is the risk tolerance for the Crandall Steel pension plan, and what should Rittenhouse recommend for the plan's allocation to cash and U.S.

equities, respectively?

	Risk Tolerance	Cash Allocation	U.S. Equities Allocation	
A)	Below average	Lower	Lower	✗
B)	Below average	Higher	Higher	✓
C)	Above average	Adequate	Higher	✗

Explanation

Given the older average age of the workforce, the high percentage of retired lives, and the fact that the plan is slightly underfunded, the risk tolerance of the plan is below average. Because the percentage of retired employees is so large, the plan is likely to have high liquidity needs. Only having 3% in cash is not likely to give the plan the liquidity it needs (the fact that the plan is underfunded is a clue here), so the allocation to cash should be higher. The plan has an actuarial required return of 6.5%. Although the plan needs to rely heavily on fixed-income instruments given the average age of its workforce and large percentage of retired lives, relying entirely on fixed income will not generate the long-term returns that the plan needs because it is a going concern. The plan should increase its allocation to U.S. equities—perhaps by reducing the allocation to venture capital because U.S. equities is a less risky asset class, but it still exceeds the plan's required return requirement.

(Module 4.6, LOS 4.k)

Question #14 of 60

Question ID: 1622915

An institutional investor has a significant allocation to private equity in its portfolio. Which of the following approaches is *least appropriate* for addressing the liquidity risk of private equity in a mean-variance optimization (MVO) context?

A) Include private equity as an asset class when running MVO, but use the risk characteristics of that specific private equity investment instead of private equity as an asset class. ✗

B) Include private equity as an asset class when running MVO using highly diversified asset class inputs. ✗

C) Only exclude private equity as an asset class when running MVO. ✓

Explanation

Illiquid asset classes (not limited to private equity only) can either be included in or excluded from MVO. If they are excluded from MVO, they should be used to meet separately set target asset allocations. The other statements are correct.

(Module 4.4, LOS 4.f)

Question #15 of 60

Question ID: 1551725

A bank is *most likely* to use which of the following approaches to liability-relative asset allocation?

A) Integrated asset-liability approach. ✓

B) Two-portfolio approach. ✗

C) Surplus efficient frontier approach. ✗

Explanation

Banks (along with insurance companies and hedge funds with short positions) make decisions about the composition of their liabilities jointly with their asset allocation decisions, which makes the integrated approach most appropriate.

Surplus optimization and the two-portfolio approach are distinct in that the composition of the liabilities is already in place when the asset allocations decisions are made, so the two decisions are made independently.

(Module 4.6, LOS 4.l)

Question #16 - 19 of 60

Question ID: 1551708

Which asset class should be increased to improve the return/risk optimization of the Takahashi portfolio?

A) International equities. ✓

B) Domestic equities. ✗

C) Domestic fixed income. ✗

Explanation

Increase the asset class with the highest risk-adjusted return (i.e., the highest excess return as a percentage of its marginal contribution to risk):

Marginal contribution to risk (MCTR) = beta × $\sigma_{portfolio}$

Domestic fixed income = 0.359 × 0.1002 = 0.0359718

Domestic equity = 1.447 × 0.1002 = 0.1449894

International equity = 1.404 × 0.1002 = 0.1406808

Excess return / MCTR:

Domestic fixed income = 3.25% – 2.50% / 0.0359718 = 20.85

Domestic equity = 6.97% – 2.50% / 0.1449894 = 30.83

International equity = 7.83% – 2.50% / 0.1406808 = **37.89**

Increase the allocation to **international equities**, as it has the highest risk-adjusted return.

(Module 4.5, LOS 4.g)

Question #17 - 19 of 60

Question ID: 1551709

Which asset class should be reduced to move the Takahashi allocation closer to the optimal mix?

A) Domestic fixed income.

B) Domestic equities.

C) International equities.

Explanation

Decrease the asset class with the lowest risk-adjusted return (i.e., excess return as a percentage of its marginal contribution to risk).

Marginal contribution to risk (MCTR) = beta × $\sigma_{portfolio}$

Domestic fixed income = 0.359 × 0.1002 = 0.0359718

Domestic equity = 1.447 × 0.1002 = 0.1449894

International equity = 1.404 × 0.1002 = 0.1406808

Excess return / MCTR:

Domestic fixed income = 3.25% – 2.50% / 0.0359718 = **20.85**

Domestic equity = 6.97% – 2.50% / 0.1449894 = 30.83

International equity = 7.83% – 2.50% / 0.1406808 = 37.89

Decrease the allocation to **domestic fixed income**, as it has the lowest risk-adjusted return.

(Module 4.5, LOS 4.g)

Question #18 - 19 of 60

Question ID: 1551710

Which asset class of the Takahashi portfolio is *closest* to its optimal weighting?

A) Domestic equities.

B) Domestic fixed income.

C) International equities.

Explanation

The portfolio is optimized if the excess return / MCTR of an allocation equals the portfolio Sharpe ratio:

Portfolio return = 0.40 (3.25%) + 0.32 (6.97%) + 0.28 (7.83%)

= 1.3% + 2.2304% + 2.1924%

= 5.72%

Portfolio Sharpe ratio = (0.0572 – 0.025) / 0.1002 = **0.3214**

Marginal contribution to risk (MCTR) = beta × $\sigma_{portfolio}$

Domestic fixed income = 0.359 × 0.1002 = 0.0359718

Domestic equity = 1.447 × 0.1002 = 0.1449894

International equity = 1.404 × 0.1002 = 0.1406808

Excess return / MCTR:

Domestic fixed income = 3.25% – 2.50% / 0.0359718 = 20.85

Domestic equity = 6.97% – 2.50% / 0.1449894 = **30.83**

International equity = 7.83% – 2.50% / 0.1406808 = 37.89

Domestic equities is closest to its optimal allocation (i.e., excess return / MCTR of 30.83 is closest to the portfolio Sharpe ratio of 0.3214).

(Portfolio Sharpe ratio = 5.72% – 2.50% / 0.1002 = 32.14)

(Module 4.5, LOS 4.g)

Question #19 - 19 of 60

Question ID: 1551711

Which of the following is an estimate of the percentage contribution of domestic bonds to overall portfolio risk of the Takahashi portfolio?

A) 13.36%.

B) 12.36%.

C) 14.36%.

Explanation

Absolute contribution to risk (ACTR) = marginal contribution to risk (MCTR) × weight:

Domestic fixed income = 0.0359718 × 0.40 = **0.01438872**

Domestic equity = 0.1449894 × 0.32 = 0.04639661

International equity = 0.1406808 × 0.28 = 0.03939062

Portfolio standard deviation = 0.1002

Domestic fixed income contributes 0.01438872 / 0.1002 = **14.36%** of the portfolio risk of 10.02%.

(Module 4.5, LOS 4.g)

Question #20 - 23 of 60

Question ID: 1551703

Based upon the data in **Exhibit 1: Selected Information Pertaining to Three Asset Allocations and Brown's Objective Function** and the safety-first ratio, which of the following is *most likely* the optimal asset allocation that has the highest chance of meeting Brown's return goal?

A) Allocation C.

B) Allocation B.

C) Allocation A.

Explanation

Brown's optimal asset allocation should meet his return goal and maximize his utility.

First, using the financial calculator, derive the annual return that will increase Brown's $550,000 savings by 25% in 10 years, subject to annual withdrawals of $20,000. Input the following: N = 10; PV = –550,000; PMT = 20,000; and FV = 550,000 × (1 + 25%) = 687,500. Next, press CPT and I/Y; the result is annual return = 5.57%.

Allocation B and Allocation C have expected returns higher than 5.57%, while Allocation A's expected return is only 5.22%.

Next, using Brown's objective function, calculate Brown's utility from Allocation B and Allocation C:

$$U_B = 6.02 - 0.02 \times 12.00^2 = 3.14\%$$

$$U_C = 6.52 - 0.02 \times 13.00^2 = 3.14\%$$

Because Allocation B and Allocation C have the same utility value, to decide which asset allocation has a higher chance of meeting Brown's goal, calculate the safety-first ratios for both allocations as

$$\text{SFR} = \frac{\text{E}(\text{R}_\text{P})\text{-R}_\text{L}}{\sigma_\text{P}}$$

where R_p is the expected return on an asset mix portfolio; R_L is the investor's return objective; and σ_p is the expected variance of return for the asset mix portfolio:

$$\text{SFR}_\text{B} = \frac{6.02\%\text{-}5.57\%}{12.00\%} = 0.0375$$

$$\text{SFR}_C = \frac{6.52\%\text{-}5.57\%}{13.00\%} = 0.0731$$

Thus, Allocation C is the optimal asset allocation because it has a higher likelihood of achieving Brown's goal.

Although Allocation A has the highest utility ($U_A = 5.22 - 0.02 \times 9.2^2 = 3.527\%$) among the three asset allocations, Brown's objective is to earn an annual return of 5.57%, which is higher than the expected return of Allocation A.

Although Allocation B has the same utility as Allocation C, it has a lower chance of achieving Brown's goal based on the safety-first ratio.

(Module 4.3, LOS 4.b)

Question #21 - 23 of 60

Question ID: 1551704

Based on **Exhibit 2: Selected Information Pertaining to Brown's Total Net Worth (USD)** and Brown's occupation, which of the following is *most likely* correct?

A) Only Suggestion 2 is recommended.

B) Only Suggestion 1 is recommended.

C) Neither suggestion is recommended.

Explanation

Suggestion 1 is not recommended. Taking into account nontradable assets may lead to improved asset allocation decisions and may increase the investor's ability to bear risk. Miller is, therefore, recommended to include the value of both Brown's human capital and inherited home when optimally allocating his total assets. Two constraints need to be added in this regard: (1) set the human capital allocation to $\frac{450{,}000}{1{,}250{,}000} = 36\%$ and (2) set the residential real estate value allocation to $\frac{250{,}000}{1{,}250{,}000} = 20\%$. Jones recommended to only include human capital, resulting in a human capital allocation constraint of $\frac{450{,}000}{1{,}000{,}000} = 45\%$.

Suggestion 2 is also not recommended. Brown is an entrepreneur running a start-up company that offers commission-based staffing services in the United States. This implies that Brown's job is not stable and his USD-based income is variable. Brown's human capital can therefore be modeled as a mix of U.S. equities, high-yield corporate bonds, and inflation-linked bonds. Thus, to limit the correlation risk of Brown's financial assets with the modeled human capital, a decrease in funds allocated to both U.S. equities and corporate bonds is recommended.

(Module 4.4, LOS 4.c)

Question #22 - 23 of 60

Question ID: 1551705

Regarding the reverse optimization and resampling techniques, which of the following is *most accurate*?

A) Both statements are correct.

B) Only Statement 2 is correct.

C) Only Statement 1 is correct.

Explanation

Statement 1 is incorrect. Both reverse optimization and resampling improve the quality of MVO inputs and lower the concentration in specific asset classes.

- *Reverse optimization* improves the quality of expected returns that are inputted into a forward-looking optimization.
- *Resampling* uses Monte Carlo simulation to estimate a large number of potential capital market assumptions for MVO. These assumptions lead to an equal number of simulated frontiers. The resulting asset allocations from the simulated frontiers are averaged and coupled with the starting capital market assumptions to draw the resampled frontier.

Statement 2 is correct. Reverse optimization works in the opposite direction of MVO. It uses as inputs a set of asset allocation weights (that are assumed to be optimal) in addition to the covariances and the risk aversion coefficient to derive the expected returns, which are then used as inputs in a forward-looking optimization. When applying reverse optimization, the market-capitalization values of the asset classes that form the diversified global market portfolio are often used to determine the starting optimal weights.

(Module 4.1, LOS 4.a)

Question #23 - 23 of 60

Question ID: 1551706

Regarding the comments given by Jones, and using **Exhibit 3: Illustrative Example About Risk Budgeting**, which of the following is *most accurate*?

A) Both comments are correct. ✗

B) Neither comment is correct. ✗

C) Only Comment 1 is correct. ✓

Explanation

Comment 1 is correct. A strategic asset allocation is said to be optimal from a risk-budgeting perspective when the excess return above the risk-free rate over the marginal contribution to total risk $\left(\frac{\mathrm{E(R_i)\text{-}R_f}}{\mathrm{MCTR_i}}\right)$ is equal for all assets and matches the optimal portfolio's Sharpe ratio. The MCTR reflects the change in the portfolio's total risk for small changes in the allocation to a specific asset class and is calculated as

$\mathrm{MCTR_i}$ = asset beta relative to portfolio × portfolio standard deviation

where *portfolio standard deviation* is the sum of the absolute contribution to total risk (ACTR) of all asset classes.

Calculate the MCTR for all asset classes:

$\mathrm{MCTR_{U.S.equities}}$ = 1.29 × 0.24 = 0.3096

$\mathrm{MCTR_{non\text{-}U.S.equities}}$ = 1.16 × 0.24 = 0.2784

$\mathrm{MCTR_{investment\text{-}grade\ bonds}}$ = 0.46 × 0.24 = 0.1104

$\mathrm{MCTR_{high\text{-}yield\ corporate\ bonds}}$ = 1.24 × 0.24 = 0.2976

Calculate the excess return above the risk-free rate over the MCTR ratio for all asset classes, where the risk-free rate is 2.25%, which is the return earned on cash:

For U.S. equities: $\frac{5.64\%-2.25\%}{0.3096}=0.109$

For non-U.S. equities: $\frac{6.01\%-2.25\%}{0.2784}=0.135$

For investment-grade bonds: $\frac{2.65\%-2.25\%}{0.1104}=0.0362$

For high-yield corporate bonds: $\frac{4.64\%-2.25\%}{0.2976}=0.0803$

Thus, the asset allocation is non-optimal from a risk-budgeting perspective because the ratio of excess return to MCTR is not the same for all assets. Note that there is no further need to calculate the portfolio Sharpe ratio.

Comment 2 is incorrect. The portfolio Sharpe ratio is $\frac{4.73\%\text{-}2.25\%}{24\%}=0.103$. To make the strategic asset allocation closer to being optimal *from a risk-budgeting perspective*, Brown needs to allocate more to assets with an excess return to MCTR that is higher than the portfolio Sharpe ratio (i.e., to U.S. and non-U.S. equities) and allocate less to assets with an excess return to MCTR that is lower than the portfolio Sharpe ratio (i.e., to investment-grade and high-yield corporate bonds).

(Module 4.5, LOS 4.g)

Question #24 of 60

Question ID: 1580701

A portfolio comprises three asset classes: 65% equities, 30% bonds, and 5% cash. The ratio of excess returns to the marginal contribution to total risk (MCTR) is equal to the portfolio's Sharpe ratio. This implies that:

A) The $MCTR_{equities}$ is equal to the absolute contribution of total risk, or $ACTR_{equities}$. ✗

B) The 65%/30%/5% allocation is optimal from a risk-budgeting perspective. ✓

C) The sum of the MCTRs of equities and bonds is equal to total portfolio risk. ✗

Explanation

Because the ratio of excess return to $MCTR_{equities}$ is equal to the portfolio Sharpe ratio (portfolio excess return / portfolio standard deviation), it implies that the current asset allocation is appropriate.

The statement that $MCTR_{equities}$ is equal to $ACTR_{equities}$ is incorrect, because $ACTR_{equities} = weight_{equities} \times MCTR_{equities}$.

The sum of the ACTRs (of equities and bonds only, because cash has zero ACTR) is equal to total portfolio risk, or standard deviation.

(Module 4.5, LOS 4.g)

Question #25 of 60

Question ID: 1580700

A portfolio comprises 70% Canadian equities and 30% Canadian bonds and has a standard deviation of 15%. The beta of Canadian equities is 1.2, while the beta of Canadian bonds is 0.65. What is the absolute contribution of total risk (ACTR) of Canadian equities?

A) 18.0%. ✗

B) 12.6%. ✓

C) 10.5%. ✗

Explanation

The percentage contribution of total portfolio risk by Canadian equities = ACTR / total portfolio risk.

The $ACTR_{Canadian\ equities} = (weight_{Canadian\ equities}) \times (MCTR_{Canadian\ equities})$, where MCTR is the marginal contribution to total risk. $MCTR_{Canadian\ equities} = beta_{Canadian\ equities} \times$ total portfolio risk.

Therefore, the % contribution of total portfolio risk by Canadian equities = $beta_{Canadian\ equities} \times$ total portfolio risk $\times\ weight_{Canadian\ equities} = 1.2 \times 0.70 \times 15\% = 12.6\%$.

(Module 4.5, LOS 4.g)

Question #26 of 60

Question ID: 1616953

The following information is available regarding corner portfolios from an efficient frontier.

Corner Portfolio	Expected Return	Exp. Std. Dev.	Asset Class Weights			
			1	2	3	4
1	6.90%	4.60%	0.00%	12.00%	88.00%	0.00%
2	10.00%	8.64%	0.00%	15.00%	45.00%	40.00%
3	13.00%	12.50%	55.00%	0.00%	45.00%	0.00%

A foundation has a spending rate of 8%. If inflation is expected to be 3.50% annually and the cost of earning investment returns is 0.005%, which of the following represents the correct weight of one of the asset classes that will, at a minimum, satisfy the investor's goals of capital preservation in real terms to an investor with a risk aversion value of 4?

A) Asset Class 4 with a weight of 19.6%.

B) Asset Class 2 with a weight of 7.7%.

C) Asset Class 1 with a weight of 27.7%.

Explanation

Foundations and endowments normally use the additive approach and include the expense rate in the return calculation:

r = s + I + c = 8 + 3.5 + 0.005 = 11.505% or 11.51%

This portfolio would lie between Corner Portfolios 2 and 3. Let *w* denote the weight of Corner Portfolio 2; we solve for *w* in the following equation:

11.51 = (10.0)(w) + (13.00)(1 – w)

w = 0.497 = 49.7%

and 1 - w = Corner Portfolio 3 = 1 - .497 = 0.503 = 50.3%

With respect to the asset classes, the weights are then derived as follows:

Weight of Asset Class 1 = (0.497)(0%) + (0.503)(55%) = 27.7%

Weight of Asset Class 2 = (0.497)(15%) + (0.503)(0%) = 7.5%

Weight of Asset Class 3 = (0.497)(45%) + (0.503)(45%) = 45.0%

Weight of Asset Class 4 = (0.497)(40%) + (0.503)(0%) = 19.9%

(Module 4.3, LOS 4.b)

Question #27 of 60

Question ID: 1580690

Shad Reed is on the board of trustees for the Wesley Ridge World Hunger Organization. The primary role of the organization is to oversee a large endowment fund that was originally established to provide food to low-income children in the United States. The original donor of the endowment recently died and provided the fund with another $200 million from his will to broaden the scope of the fund to provide food for hungry children all over the world. With the addition, the endowment's assets are currently valued at $600 million. When the fund was originally established, the spending rate was 5%; however, with the broader scope, the spending rate has increased to 6%. Also, because funds are going to be distributed to other countries, the board has determined that approximately 25% of the foundation's annual payout will be in the foreign currencies of other countries. The fund's revised investment policy statement is shown as follows:

Return Objective	Accounting for inflation of 2.5% and the new spending rate of 6%, the return requirement for the plan is 8.5%. A total return approach is appropriate.
Risk Tolerance	Above average, although risk tolerance has declined due to higher spending needs.
Liquidity	The endowment has minimal operating expenditures—liquidity requirements are low.
Time Horizon	Long-term
Legal/Regulatory	N/A
Taxes	N/A
Unique Considerations	N/A

The board has consulted with an investment advisor to discuss changes to the endowment's current asset allocation, which is shown as follows:

Asset Class	Allocation (%)	Expected Return	Expected Standard Deviation
Cash and marketable securities	7%	3.0%	2%
Intermediate-term U.S. Treasury bonds	28%	5.5%	7%
Foreign government bonds	8%	6.5%	10%
U.S. equities	45%	9.5%	18%
International equities	7%	11.0%	23%

Venture capital	5%	19.0%	38%

Which of the following recommendations would be *most appropriate* for the endowment fund?

A) Increase the allocation to foreign government bonds, increase the allocation to international equities, keep the allocation to cash the same, and keep the allocation to venture capital the same. ✓

B) Decrease the allocation to U.S. equities, decrease the allocation to international equities, increase the allocation to foreign government bonds, and increase the allocation to intermediate-term U.S. Treasury bonds. ✗

C) Increase the allocation to cash, decrease the allocation to U.S. equities, decrease the allocation to international equities, and increase the allocation to venture capital. ✗

Explanation

Because liquidity needs are low, the allocation to cash is appropriate (because the 7% allocation slightly exceeds the 6% annual spending rate). Because payments are going to be made in foreign currencies, it makes sense to increase the allocation to both foreign bonds and international equities. Especially if these asset classes have low correlation with other assets in the portfolio, increasing their weight in the portfolio could reduce risk, although their higher standard deviations suggest only moderate increases. Because the fund has a long time horizon and above-average risk tolerance, the allocation to venture capital is fine given its above-average returns and likely low correlation with other assets in the portfolio.

(Module 4.3, LOS 4.d)

Question #28 of 60

Question ID: 1580687

The following information is a partial list of corner portfolios and asset class weights:

Portfolio	Exp. Return	Std. Dev.	Asset Class Weights		
			1	2	3
1	12.00%	10.50%	65.00%	0.00%	35.00%
2	16.50%	14.00%	15.00%	20.00%	50.00%
3	18.00%	20.00%	30.00%	20.00%	25.00%
4	23.00%	24.00%	15.00%	20.00%	55.00%

Which asset class is the most significant for an efficient portfolio with an expected return of 15% and the approximate standard deviation of this efficient portfolio?

A) Asset Class 1, 15.00%.

B) Asset Class 3, 12.85%.

C) Asset Class 2, 12.85%.

Explanation

The expected return of 15% lies between Corner Portfolios 1 and 2 with expected returns of 12% and 16.50%. We solve for *w* in the following equation:

$$15 = w(12) + (1 - w)(16.50)$$

$$w = 0.33$$

In other words, the efficient portfolio with an expected return of 15% has 33% weight of Corner Portfolio 1 and 67% weight of Corner Portfolio 2. With respect to the asset classes, the weights are then derived as follows:

Weight of Asset Class 1 = (0.33)(65%) + (0.67)(15%) = 31.50%

Weight of Asset Class 2 = (0.33)(0%) + (0.67)(20%) = 13.4%

Weight of Asset Class 3 = (0.33)(35%) + (0.67)(50%) = 45.05%

Asset Class 3 has the highest weight and is the most significant.

Approximate standard deviation = (0.33)(10.50) + (0.67)(14) = 12.85%.

(Module 4.3, LOS 4.b)

Question #29 of 60

Question ID: 1551733

Which of the following factors would *most likely* result in a narrower rebalancing corridor?

A) Higher transaction costs.

B) Higher volatility of asset classes.

C) Higher investor risk tolerance.

Explanation

High asset class volatility would result in a narrower rebalancing corridor because higher volatility increases the need for quicker rebalancing as the asset mix moves away from the target allocation.

Higher investor risk tolerance allows for a *wider* rebalancing corridor because the investor is now less concerned with deviations of the asset mix away from the target allocation.

Higher transaction costs allows for a *wider* rebalancing corridor because the asset mix would have to change more to justify the higher cost of rebalancing.

(Module 4.8, LOS 4.o)

Question #30 of 60

Question ID: 1580699

Which of the following statements regarding risk and risk budgeting is correct?

A) Active risk is most relevant in an asset allocation implementation setting.

B) For risk budgeting purposes, risk can be defined as total risk, active risk, or misfit risk.

C) An optimal risk budget minimizes the total amount of portfolio risk that is allocated to the portfolio's constituent parts.

Explanation

Overall market risk is relevant in an asset allocation setting, and active risk is relevant in an asset allocation implementation setting.

An optimal risk budget allocates risk efficiently in that it attempts to maximize the return per unit of risk taken.

For risk budgeting purposes, risk can be defined as total risk, active risk, or residual risk.

(Module 4.5, LOS 4.g)

Question #31 of 60

Question ID: 1580692

Aaron Manning wishes to minimize the risk of his portfolio returns as measured by standard deviation while meeting a minimum expected return objective. The risk-free rate is 2%. The following asset allocations are available:

	Expected Return	Standard Deviation of Returns
Allocation 1	10%	12%
Allocation 2	7%	10%
Allocation 3	11%	15%

Based on the information provided, which of the following allocations should Manning choose?

A) Allocation 3.

B) Allocation 1.

C) Allocation 2.

Explanation

Calculate the Sharpe ratio for each allocation:

Allocation 1: (10% – 2%) / 12% = 0.67

Allocation 2: (7% – 2%) / 10% = 0.50

Allocation 3: (11% – 2%) / 15% = 0.60

Allocation 1 has the highest Sharpe ratio; therefore, it is the optimal risk allocation that Manning should choose.

(Module 4.3, LOS 4.d)

Question #32 of 60

Question ID: 1587529

A 30-year-old wealthy investor wants to maximize her allocation to commodity investments. Which of the following approaches to asset allocation would *best* achieve her objective?

A) Endowment model.

B) 120 minus age rule.

C) 1/N rule, with 10 asset classes.

Explanation

The endowment model (or Yale model) allows for higher allocation to alternative investments, including commodities, real estate, and private equity, than recommended under mean-variance optimization (MVO). The investor should select managers with significant exposure to these alternative asset classes. The model does not cap the allocation to alternative investments.

The 120 minus age rule considers only two asset classes: equities and fixed income, with the equity allocation percentage determined as 120 minus age, but it does not consider alternative investments.

The 1/N rule considers an equally weighted portfolio to each selected asset class. If commodities is one of the 10 asset classes selected, then this approach caps the commodities allocation at 1/10 = 0.10, or 10%, which may be insufficient for the investor.

(Module 4.7, LOS 4.n)

Question #33 of 60

Question ID: 1587527

Which of the following statements about an investor's goals-based asset allocation approach is *least correct*?

A) The investor's different goals with specified probabilities of success will be averaged, and a single module with the highest expected return will be used for the asset allocation. ✓

B) The goals-based asset allocation approach typically uses pre-established subportfolios to meet client goals. ✗

C) The asset opportunity sets will consist of both taxable and tax-exempt investments. ✗

Explanation

The advantage of the goals-based asset allocation approach is that it is able to incorporate different goals by an investor that could differ by amount, timing and urgency. Each goal is evaluated separately by looking at modules (pre-established subportfolios) that provide the minimum expected return needed to satisfy that goal. Goals are not averaged, and multiple modules are used because each module will result in different minimum returns depending on the level of specified probabilities of success.

The asset opportunity set for each module consists of both taxable and tax-exempt investments.

(Module 4.7, LOS 4.m)

Question #34 of 60

Question ID: 1587520

Mean-variance optimization (MVO) outputs using factor exposures differ from using asset class exposures that reflect the same underlying assets in that:

A) their efficient frontiers differ from each other. ✗

B) the choice of exposures reflects how an investor forms capital market expectations. ✓

C) one approach will typically be superior to the other, depending on the space in which the investor operates. ✗

Explanation

The choice of whether to use factor exposures or asset class exposures depends on the way investors form capital market expectations.

When factor exposures and asset class exposures reflect the same exposures (i.e., underlying assets), the MVO results indicate that neither approach is superior to the other. As a result, their efficient frontiers will not be significantly different.

(Module 4.5, LOS 4.i)

Question #35 of 60

Question ID: 1580697

An institutional portfolio contains the following asset classes: direct real estate, infrastructure, and private equity. Which of the following statements is correct regarding the potential inclusion of those asset classes in performing mean-variance optimization (MVO)?

A) There are numerous indices available to accurately track such investments. ✗

B) The risk-return characteristics of a specific investment in those assets are typically similar to the risk-return characteristics of the asset class. ✗

C) Indices that track such investments are typically not investable. ✓

Explanation

Indices that track less liquid asset classes (i.e., direct real estate, infrastructure, private equity) are typically not investable as a passive alternative to active management. Therefore, they are difficult to include in MVO.

There are *very few* indices that accurately track infrastructure investments, which makes it harder to find data to use for estimating return, risk, and correlations for inclusion in MVO.

Specific investments in less liquid assets are not fully diversified, so they have both systematic and unsystematic risks. In contrast, asset classes tend to be broader and more diversified, thereby likely reflecting very little unsystematic risk. Therefore, the return-risk characteristics of specific illiquid assets and their corresponding asset classes are *not similar*, which makes it difficult to include such asset classes in MVO.

(Module 4.4, LOS 4.f)

Question #36 of 60

Question ID: 1587525

Taylor Robinson, age 60, recently retired from her position as director of public giving for United Electric Power, a large public utility company. Robinson has accumulated $2,000,000 in her 401(k) portfolio for retirement. Robinson estimates that she will need $50,000 after tax in today's dollars to live comfortably. Inflation is expected to be 2.5% annually. With her background in public giving, Robinson has two favorite charities and would like to make non-tax-deductible gifts of $10,000 to each of them annually, indexed for inflation. In her will, Robinson has specified that at her death, a gift fund will be established for each charity. Given this objective, one of Robinson's primary goals is to maintain the principal in her retirement fund to have a $1,000,000 gift account for each charity. Robinson recently met with her financial advisor, Brian Mitchell, CFA. During their meeting, Robinson stated, "If I wanted to gamble with my investments, I would play blackjack. At least then I would have fun losing money." Mitchell presented Robinson with three different model portfolios.

Asset Class	Portfolio A	Portfolio B	Portfolio C
Large-cap U.S. stocks	5%	15%	25%
Small-cap U.S. stocks	5%		15%
International—developed market equities	10%	5%	10%
International—emerging market equities	15%		5%
U.S. corporate bonds	10%	50%	30%
U.S. Treasury bonds	5%	20%	
Hedge fund of funds	10%		5%
Venture capital	15%		
Cash	25%	10%	10%
Total expected after-tax return	5.8%	5.5%	6.0%
Current yield	1.4%	1.6%	1.8%

Which of the portfolios would be *most appropriate* for Robinson?

A) Portfolio B.

B) Portfolio A.

C) Portfolio C.

Explanation

Based on the information given, it would appear that Portfolio C would be the best choice. Robinson needs $50,000 per year to meet her living expenses with an additional $20,000 per year given to charity. This implies a real required return of $70,000 / $2,000,000 = 3.5% after taxes. Factoring in inflation = 3.5 + 2.5 = 6.00% or (1.035)(1.025) – 1 = 6.09%. Robinson's statement about not wanting to gamble with her investments implies a low risk tolerance. Portfolio C provides a return of at least the required 6.0%, and it has a respectable income component given the other choices. It also appears to be well diversified among various asset classes. The small allocation to a fund of hedge funds provides an asset with low correlation that should reduce risk. Portfolio B is too conservative and does not meet Robinson's return requirement. Portfolio A looks like a good choice because its return is close to the required return, but it has a high allocation to cash. If we take a closer look at the allocation of the portfolio, it would seem the 15% venture capital allocation is likely driving Portfolio A's return. Because Portfolio A falls short of the required return and has a high cash allocation and 15% allocated to venture capital, Portfolio A would not be the best choice.

(Module 4.7, LOS 4.m)

Question #37 of 60

Question ID: 1587528

Which of the following heuristic and other approaches to asset allocation is *most closely* associated with the assumption of a lack of informationally efficient markets?

A) Norway model.

B) Endowment model.

C) 60/40 stock/bond heuristic model.

Explanation

The endowment model has large allocations to alternative assets as well as support for active management. The endowment model also seeks to earn illiquidity premiums. Those three factors suggest that there is the assumption of a lack of informationally efficient markets.

The Norway model's asset allocation emphasizes publicly traded securities, which reflects a belief in the market's informational efficiency.

(Module 4.7, LOS 4.n)

Question #38 of 60

Question ID: 1587531

In context of portfolio rebalancing, if the correlation of the asset class with the rest of the portfolio is lower, then the optimal corridor of the asset class will be:

A) narrower. ✓

B) unchanged. ✗

C) wider. ✗

Explanation

The lower the correlation of the asset class with the rest of the portfolio, the narrower the corridor, because the portfolio will not "move" as closely with the asset class—so the allocations are more likely to diverge from the target allocation. Therefore, the corridor needs to be narrowed to control the risk.

(Module 4.8, LOS 4.o)

Question #39 of 60

Question ID: 1580682

Jeff Graefe has a risk aversion value of 6. He is evaluating three competing investments with the following characteristics. Which investment would have the *least* utility for Graefe?

Portfolio	Return	Std. Dev.
A	18.0%	24.0%
B	13.5%	10.0%
C	9.5%	6.0%

A) B. ✗

B) A. ✓

C) C. ✗

Explanation

$R_Z = 6$

$U_A = E(R_A) - 0.5(R_Z)(\sigma^2_A) = 0.18 - 0.5(6)(0.24)^2 = 0.007$

$U_B = E(R_C) - 0.5(R_Z)(\sigma^2_C) = 0.135 - 0.5(6)(0.10)^2 = 0.105$

$U_C = E(R_D) - 0.5(R_Z)(\sigma^2_D) = 0.095 - 0.5(6)(0.06)^2 = 0.084$

Investment A has the least utility of 0.007.

(Module 4.1, LOS 4.a)

Question #40 of 60

Question ID: 1580683

Dan Laske is evaluating three portfolios for investment of his retirement funds. Laske has a risk aversion value of 5. Which portfolio would be *best* for him?

Portfolio	Return	Std. Dev.
A	15.0%	17.0%
B	10.6%	10.0%
C	8.8%	8.0%

A) B. ✓

B) A. ✗

C) C. ✗

Explanation

$R_Z = 5$

$U_A = E(R_A) - 0.5(R_Z)(\sigma^2_A) = 0.15 - 0.5(5)(0.17)^2 = 0.078$

$U_B = E(R_C) - 0.5(R_Z)(\sigma^2_C) = 0.106 - 0.5(5)(0.10)^2 = 0.081$

$U_C = E(R_D) - 0.5(R_Z)(\sigma^2_D) = 0.088 - 0.5(5)(0.08)^2 = 0.072$

Portfolio B has the highest utility.

(Module 4.1, LOS 4.a)

Question #41 of 60

Question ID: 1587526

Which of the following statements regarding the goals-based approach to asset allocation is correct?

A) The relevant risk measure for both individuals and institutions is the probability of missing the goal. ✗

B) Individuals are often concerned with single goals, while institutions are often concerned with multiple goals. ✗

C) The return determination for individuals is based on minimum expectations, while for institutions, it is based on mathematical expectations. ✓

Explanation

For institutions, *mathematical expectations* refers to the weighted expected return of the portfolio components.

The relevant measure for institutions is volatility of return or surplus.

Individuals are often concerned with *multiple* goals, while institutions are often concerned with *single* goals.

(Module 4.7, LOS 4.m)

Question #42 of 60

Question ID: 1587524

A risk manager at HIJ Bank is worried that an increase in interest rates will negatively impact the bank's surplus, while an increase in risk volatility may negatively impact some of its liabilities. Which liability-relative asset allocation approach would be *most appropriate* for the bank?

A) Two-portfolio approach. ✗

B) Surplus optimization. ✗

C) Integrated asset-liability approach. ✓

Explanation

Because the bank wants to look at both sides of the balance sheet through stress testing against future changes in multiple factors (interest rates and risk volatility), an integrated asset-liability approach is most appropriate. That approach will allow the bank to determine the optimal asset and liability mix to meet its objectives.

Surplus optimization is an extension of mean-variance optimization (MVO) that manages the portfolio surplus against the surplus volatility.

The two-portfolio approach segments the portfolio assets to hedge liabilities, and it separately manages the remaining portfolio using MVO.

(Module 4.6, LOS 4.l)

Question #43 of 60

Question ID: 1580689

David Jalbert is considering three potential asset allocations. He wishes to earn a nominal return no less than 2% as he has a low risk tolerance with a lambda of 7.

The following asset allocations are available:

	Expected Return	Variance
Allocation 1	5%	0.0100
Allocation 2	7%	0.0143
Allocation 3	9%	0.0200

Based on the information provided, which of the following allocations should Jalbert choose?

A) Allocation 1.

B) Allocation 2.

C) Allocation 3.

Explanation

Step 1: Calculate the certainty-equivalent return.

Allocation 1: 0.05 – (0.5 × 7 × 0.01) = 0.01500

Allocation 2: 0.07 – (0.5 × 7 × 0.0143) = 0.01995

Allocation 3: 0.09 – (0.5 × 7 × 0.02) = 0.02000

Jalbert would probably be indifferent between Allocations 2 and 3 based on the certainty-equivalent return and prefers them both to Allocation 1.

Step 2: Calculate the Roy's safety-first criterion.

Allocation 1: $(0.05 - 0.02) / (0.0100)^{1/2} = 0.300$

Allocation 2: $(0.07 - 0.02) / (0.0143)^{1/2} = 0.418$

Allocation 3: $(0.09 - 0.02) / (0.0200)^{1/2} = 0.495$

Allocation 3 has a higher probability of exceeding the threshold return than Allocation 2, so Jalbert should choose Allocation 3.

(Module 4.3, LOS 4.b)

Question #44 - 47 of 60

Question ID: 1551688

For Elias Weber, for Portfolios B, D, and F, which of the following are ranked from highest to lowest in their utility (certainty equivalent return)?

A) F, B, D. ✗

B) B, D, F. ✓

C) D, F, B. ✗

Explanation

Utility = E/R – (0.005 × λ × σ^2)

B = 8.76 – (0.005 × 4.5 × 14.78^2) = **3.845**

D = 7.15 – (0.005 × 4.5 × 12.64^2) = 3.555

F = 5.25 – (0.005 × 4.5 × 10.11^2) = 2.950

Allocation B has the highest utility, followed by D, then F.

Candidate discussion: To use decimals, change 0.005 to 0.5 as follows:

= 0.0876 – (0.5 × 4.5 × 0.1478^2) = **3.845**.

(Module 4.3, LOS 4.d)

Question #45 - 47 of 60

Question ID: 1551689

For Lina Wagner, assuming Allocation A is recommended, and borrowing and lending at the risk-free rate is permitted, which of the following is the correct allocation to the risk-free rate?

A) EUR 7.60 million. ✓

B) EUR 6.80 million. ✗

C) EUR 7.40 million. ✗

Explanation

Solve to find the required weight in Allocation A:

$$5.5 = 9.35w + 1 - w(1.75)$$

$$5.5 = 9.35w + 1.75 - 1.75w$$

$$3.75 = 7.6w$$

$$w = 3.75 / 7.6$$

$$w = 0.4934$$

In other words: 49.34% in Allocation A, and **50.66% in risk free**

$$0.5066 \times \text{EUR } 15 \text{ million} = \textbf{EUR 7,599,000 in risk free}$$

To check:

$$(0.4934 \times 9.35\%) + (0.5066 \times 1.75\%) = 5.50\%$$

(Module 4.3, LOS 4.d)

Question #46 - 47 of 60

Question ID: 1551690

For Tobias Weisser to achieve his required return of 7.5%, which of the following is *most correct*?

A) Lend EUR 0.412 million at the risk-free rate and invest EUR 7.412 million in Allocation D. ✗

B) Borrow EUR 0.412 million at the risk-free rate and invest EUR 7.412 million in Allocation D. ✗

C) Borrow EUR 0.454 million at the risk-free rate and invest EUR 7.454 million in Allocation D. ✓

Explanation

Use the most efficient allocation allowable (i.e., the one with the highest Sharpe ratio—Portfolio D and the risk-free asset):

A = (9.35 – 1.75) / 16.00 = 0.475

B = (8.76 – 1.75) / 14.78 = 0.474

C = (8.24 – 1.75) / 14.25 = 0.455

D = (7.15 – 1.75) / 12.64 = **0.427**

E = (5.50 – 1.75) / 10.32 = 0.363

F = (5.25 – 1.75) / 10.11 = 0.346

G = (3.60 – 1.75) / 7.28 = 0.254

Solve to find the weight in Allocation D:

7.50 = 7.15w + 1.75 (1 – w)

7.50 = 7.15w + 1.75 – 1.75w

7.50 – 1.75 = 7.15w – 1.75w

5.75 = 5.40w

w = 1.0648

So: 106.481% in Allocation D and –6.481% in risk free

To check:

(1.06481 × 7.15%) + (–0.06481 × 1.75%) = **7.50%**

In Allocation D: EUR 7 million × 1.06481 = **EUR 7.454 million**

Borrow at the risk free: **EUR 0.454m**

(Module 4.3, LOS 4.d)

Question #47 - 47 of 60

Question ID: 1551691

Hoffman meets again with Lina Wagner, who advises that she now requires a minimum return of at least 2.5%, and a maximum volatility of 13%. Which of the following allocations should now be recommended to Wagner by Hoffman?

A) Allocation A. ✗

B) Allocation E. ✗

C) Allocation D. ✓

Explanation

Using Roy's safety-first approach, with the additional constraint of a maximum of 13% volatility:

A = 9.35 – 2.50 / 16.00 = 0.428

B = 8.76 – 2.50 / 14.78 = 0.423

C = 8.24 – 2.50 / 14.25 = 0.403

D = 7.15 – 2.50 / 12.64 = **0.368**

E = 5.50 – 2.50 / 10.32 = 0.291

F = 5.25 – 2.50 / 10.11 = 0.272

Allocation D has the highest Roy's safety-first *z*-score, meaning it has the lowest probability of shortfall below the minimum target return of 2.5%. Although Allocations A, B, and C have even lower probabilities of shortfall (higher RSF scores), the risk is higher than 13%, so they have been excluded.

(Module 4.3, LOS 4.d)

Question #48 of 60

Question ID: 1580695

Consider an investor whose future wages are uncertain. The investor's human capital could be modeled as a(n):

A) mix of inflation-linked bonds, equities, and corporate bonds. ✓

B) inflation-linked bond. ✗

C) real return bond. ✗

Explanation

Investors with uncertain future wages are exposed to various risks. As a result, their human capital should be modeled as a combination of various risks, including a mix of inflation-linked bonds, equities, and corporate bonds.

For investors who have jobs that are stable and increase with inflation, modeling their human capital as inflation-linked bonds is appropriate.

(Module 4.4, LOS 4.c)

Question #49 of 60

Question ID: 1587530

The highest percentage allocation to fixed income for a 25-year-old investor would be under the:

A) 60/40 split.

B) 120 minus age rule.

C) 1/N rule, with 3 asset classes.

Explanation

Under a 60/40 split, the investor's allocation to fixed income is permanently 40%, with a 60% allocation to equities, regardless of age. This approach assumes there is no difference in asset allocation preferences between investors.

Under the 120 minus age rule, a 25-year-old investor would have 120 – 25 = 95% allocation to equities and 5% allocation to fixed income.

Under the 1/N rule, the portfolio is equally weighted along each selected asset class. If fixed income is one of the three asset classes selected, then this approach caps the fixed income allocation at 1/3 = 0.33, or 33%.

(Module 4.7, LOS 4.n)

Question #50 of 60

Question ID: 1674562

A problem with mean variance optimization (MVO) is that:

A) it starts with portfolio weights from the global market portfolio, which are assumed to be optimal.

B) asset allocations are sometimes highly concentrated in a small number of asset classes while ignoring other asset classes.

C) an output is the implied returns, which must be view adjusted.

Explanation

MVO often results in highly concentrated portfolios in a limited number of asset classes while ignoring other asset classes, resulting in a portfolio that is not practical. Implied returns are an output of reverse optimization, not MVO. The implied returns from reverse optimization are then used as inputs into MVO or view adjusted according to the investor's beliefs, referred to as the Black-Litterman Model, and then used as inputs into MVO. Reverse optimization starts with the global market portfolio's weights, which are assumed to be optimal.

(Module 4.1, LOS 4.a)

Question #51 of 60

Question ID: 1674563

Reverse optimization starts with:

A) expected returns, variances, correlations, a risk aversion factor, and constraints. ✗

B) the global market portfolio's weights, which are assumed to be optimal. ✓

C) view adjusted returns, which are then used as inputs to mean variance optimization (MVO). ✗

Explanation

Reverse optimization starts with the global market portfolio's weights, which are assumed to be optimal; with implied returns as an output, which are then used as an input to (MVO). The Black-Litterman Model uses the implied returns from the output of reverse optimization and then view adjusts those returns according to the beliefs of the investor, with the view adjusted returns used as inputs to MVO. MVO starts with inputs of: expected returns, variances, correlations, a risk aversion factor, and constraints.

(Module 4.1, LOS 4.a)

Question #52 of 60

Question ID: 1580694

Andrew Tyson, age 31, has an existing investment portfolio consisting of investment-grade bonds and large-cap stocks. He also owns an apartment unit in the city and has a stable and promising career as an engineer. By including his apartment and his human capital in his investment portfolio, his capacity to bear risk would *most likely*:

A) decrease. ✗

B) remain unchanged. ✗

C) increase. ✓

Explanation

Because human capital and residential real estate are two large, but often overlooked components of an investor's total investment portfolio, including them in the analysis along with traditional investments would *increase* the investor's capacity to bear risk.

(Module 4.4, LOS 4.c)

Question #53 of 60

Question ID: 1551674

Any mean-variance efficient portfolio has the:

A) lowest standard deviation and the highest expected return. ✗

B) lowest standard deviation for a given level of expected return. ✓

C) highest return among all other portfolios.

Explanation

A mean-variance efficient portfolio has the lowest standard deviation for a given level of expected return. Note that the lowest standard deviation portfolio and the highest return portfolio are just two of the infinite number of efficient portfolios.

(Module 4.1, LOS 4.a)

Question #54 of 60

Question ID: 1580691

Frances Bonner, a retired nurse, is a potential new client for Fullen Capital Management (Fullen). Bonner has told Fullen that to pay the living expenses not covered by her pension and Social Security, she must generate $10,000 annually on her $500,000 investment portfolio. She does not want to use principal to meet her living expenses. Fullen has four model portfolios that Bonner could use for her portfolio.

Portfolio	Expected Return	Standard Deviation
A	4.5%	6%
B	5.5%	8%
C	6.5%	10%

Based on Roy's safety-first measure, Bonner should select:

A) Portfolio A.

B) Portfolio B.

C) Portfolio C.

Explanation

Bonner states that she requires $10,000 annually on a $500,000, which implies a minimum return of $10,000 / $500,000 = 0.02 = 2%.

Roy's safety-first measure is calculated as: $SF = (R_P - R_{MIN}) / \sigma_P$

$SF_A = (4.5 - 2) / 6 = 0.4167$

$SF_B = (5.5 - 2) / 8 = 0.4375$

$SF_C = (6.5 - 2) / 10 = 0.4500$

Because Portfolio C has the highest Roy's safety-first measure, Bonner should select Portfolio C.

(Module 4.3, LOS 4.d)

Question #55 of 60

Question ID: 1587517

Corri Morgan is an investment advisor for Izaguirre Investment Management (IIM). Morgan is reviewing the account for Brian and Nicole Herbster.

Brian and Nicole are both age 65, and have one daughter, Andrea, age 18. The Herbsters are recently retired from Tucker Technology, Inc., a large manufacturer of microprocessors for various applications. Andrea is an aspiring finance student and would like to attend a prestigious university to pursue a degree in finance. The tuition at the university costs $40,000 per year, but Andrea's strong academic performance in high school allowed her to earn a scholarship covering half of the tuition. The Herbsters have expressed a desire to fund the amount of the college education not covered by the scholarship, as well as leave a large inheritance to Andrea at their death. During their careers, the Herbsters earned relatively high incomes, and they were able to save approximately 10% of their income each year. Regarding their portfolio, they say they prefer investments that have minimal volatility. Their current investment portfolio is valued at $1.2 million.

The investment policy statement for the Herbsters is shown here:

Return Objective	The Herbsters' income requirement is $6,000 monthly. Total return requirement is $72,000 / $1,200,000 = 6%.
Risk Tolerance	Ability to take on risk: average. Willingness to take risk: below average.
Liquidity	Need cash each year for the next four years to fund college education.
Time Horizon	Three stages. Stage 1: funding daughter's college tuition. Stage 2: retirement. Stage 3, after death: inheritance for daughter.
Legal/Regulatory	N/A
Taxes	Little need to defer income.
Unique Considerations	Desire to fund daughter's college education.

The Herbsters' current portfolio is shown here:

Asset/Fund	Allocation (%)	Expected Return	Expected Yield	Expected Standard Deviation
Tucker Technology common stock	32%	19.0%	0.5%	28%
Money market fund	2%	2.5%	2.5%	2%
Diversified bond fund	30%	6.5%	5.5%	8%

Large capitalization equities	15%	9.5%	2.0%	16%
Emerging market equities	15%	16.0%	1.0%	26%
Undeveloped commercial land	6%	19.0%	0%	N/A

After reviewing the notes on the Herbsters, Morgan reviews recommendations complied by Todd Irons, a fellow portfolio manager with IIM. Irons's recommendations include the following:

Recommendation 1:	Reduce the weighting in Tucker Technology common stock—the large position exposes the portfolio to unnecessary security-specific risk.
Recommendation 2:	Increase the allocation to the diversified bond fund to increase income and decrease volatility.
Recommendation 3:	Increase the allocation to large capitalization equities to provide growth.
Recommendation 4:	Maintain the allocation to emerging market equities due to their high returns.
Recommendation 5:	Maintain the allocation to the undeveloped commercial land due to its low correlation with other assets in the portfolio.

After reviewing Irons's recommendations, Morgan should agree with:

A) Recommendations 1, 3, and 5 only.

B) Recommendations 1, 2, and 3 only.

C) Recommendations 1, 2, and 4 only.

Explanation

Morgan should agree with Recommendations 1, 2, and 3. The allocation to emerging market equities is probably too high given the Herbsters' preference for low-volatility investments. Also, the allocation to undeveloped commercial land would be a cash drain on the Herbsters' portfolio due to payments for taxes and the like. The Herbsters need income and liquidity to meet ongoing portfolio disbursement requirements, and the undeveloped commercial land provides neither.

(Module 4.5, LOS 4.h)

Question #56 of 60

Question ID: 1551716

Which of the following statements about using investment factors in constructing asset allocations is *correct*?

A) The investment factors are market-neutral, long/short portfolios. ✗

B) The use of investment factors is consistent with factor return models. ✓

C) The factors are constructed to have high correlation with each other. ✗

Explanation

The use of investment factors is consistent with factor return models, including the three-factor Fama-French model (where the factors are size, value and market).

The investment factors are *zero-dollar*, long/short portfolios that are long the outperforming attribute and short the underperforming attribute.

The investment factors are constructed to have *low* correlation with each other and with the market portfolio, which results in superior risk-return tradeoff.

(Module 4.5, LOS 4.i)

Question #57 of 60

Question ID: 1551724

The investment committee of a life insurance company recommends a strategic asset allocation for the company based on the projected policy premium inflows and payouts along with long-term capital market expectations. This approach to strategic asset allocation is known as the:

A) asset-liability approach. ✓

B) static approach. ✗

C) investment policy statement approach. ✗

Explanation

Because the committee takes into account the company's inflows and outflows (liabilities), the approach is called the asset-liability approach to strategic asset allocation.

(Module 4.6, LOS 4.l)

Question #58 of 60

Question ID: 1580684

Which of the following statements regarding mean variance optimization (MVO) is *least accurate*?

A) All asset weights add up to 100%. ✗

B) Short positions are permitted. ✓

C) An individual with average risk aversion will have a lambda of about 4. ✗

Explanation

A common constraint in MVO is the nonnegativity constraint, which means that all weights in the portfolio are positive and between 0% and 100% (there are no short positions).

The most common constraint in MVO is known as the budget or utility constraint, which means the asset weights must add up to 100%.

Lambda is based on an investor's willingness and capacity to take on risk. In practice, investors are assumed to have a lambda between 1 and 10, with an average level of 4.

(Module 4.1, LOS 4.a)

Question #59 of 60

Question ID: 1587522

A pension plan wants to account for liability characteristics in its asset allocation decisions with respect to higher-than-expected payouts in the future. Which of the following liability characteristics *best* describes the plan's concerns?

A) Quasi-legal liabilities. ✗

B) Legal liabilities. ✗

C) Longevity risk. ✓

Explanation

Longevity risk is the risk that a pension plan would have to make higher than expected payouts in the future, which is typically due to higher life expectancy of pension plan members.

Legal liabilities are liabilities defined through legal agreements, which would include a pension plan's obligation to make pension payments. However, the focus of the question is on higher-than-expected payouts in the future—so although *legal liabilities* is a correct answer, it is not the best answer.

Quasi-legal liabilities refers to expected future cash outflows that are not due to legal obligations. A pension plan's obligations are clearly legal liabilities, not quasi-legal liabilities.

(Module 4.6, LOS 4.j)

Question #60 of 60

Question ID: 1587519

Which of the following statements regarding factor-based asset allocation is correct?

A) The factors may be highly correlated with the market portfolio. ✗

B) The size factor is the combined return from a short position in small stocks and long position in large stocks. ✗

C) The value-growth factor is the combined return from a short position in growth stocks and a long position in value stocks. ✓

Explanation

The Fama-French model is an example of factor-based asset allocation. There are three factors to consider:

- A zero-dollar portfolio long in small stocks and short in large stocks (the size factor)
- A zero-dollar portfolio long in value (high book-to-market) stocks and short growth (low book-to-market) stocks (the value-growth factor)
- The market portfolio

Because of the way the factors are formed, they are not correlated with each other or the market portfolio, which improves the risk-return tradeoff from the optimal portfolios and expands the efficient frontier.

(Module 4.5, LOS 4.i)